

BRUNSWICK, ME, USA

WMO: 743920

Lat: 43.900N

Lon: 69.933W

Elev: 21

StdP: 101.07

Time Zone: -5.00 (NAE)

Period: 94-10

WBAN: 14611

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.1	(c) -16.8	(d) -27.7	(e) 0.3	(f) -17.3	(g) -24.8	(h) 0.4	(i) -14.8	(j) 12.1	(k) 1.5	(l) 11.0	(m) -2.9	(n) 2.4	(o) 330	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	29.9	21.4	28.0	20.4	26.6	19.4	22.9	27.6	21.8	26.3	20.8	24.5	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.0	25.4	20.4	15.1	24.2	19.3	14.1	22.9	67.9	27.7	63.8	26.2	60.3	24.7	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	8.9	8.0	-24.1	33.6	3.2	1.8	-26.5	34.9	-28.4	35.9	-30.2	36.9	-32.5	38.2
			-24.5	25.2	3.1	1.2	-26.7	26.0	-28.6	26.7	-30.3	27.3	-32.6	28.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-6.1	-4.5	0.4	6.4	12.0	17.6	20.5	20.2	15.7	9.3	3.9	-1.9		
	DBStd	10.05	6.05	5.33	4.80	3.40	3.45	3.66	2.87	2.69	3.43	3.84	4.56	5.01		
	HDD10.0	1959	498	407	299	118	16	1	0	0	2	61	188	369		
	HDD18.3	4023	756	640	557	359	200	56	12	13	91	281	432	627		
	CDD10.0	1176	0	0	1	9	78	229	325	316	173	38	7	1		
	CDD18.3	197	0	0	0	0	3	34	78	69	12	1	0	0		
	CDH23.3	1363	0	0	2	3	43	264	529	440	78	3	0	0		
	CDH26.7	331	0	0	1	0	10	70	136	98	16	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.7	3.9	3.9	3.6	3.1	3.0	2.8	3.1	3.4	3.5	3.5		
(15) Precipitation	PrecAvg	1183	92	91	108	100	97	92	74	86	85	100	127	117		
	PrecMax	1711	275	186	260	226	224	231	201	248	217	270	364	254		
	PrecMin	911	13	3	19	10	14	17	15	9	15	24	31	29		
	PrecStd	204	52	40	50	40	49	49	40	52	45	56	64	63		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	9.0	15.9	22.1	27.8	31.3	32.3	32.0	28.7	23.0	17.2	13.6		
		MCWB	9.5	5.7	8.9	13.3	17.6	21.8	22.9	22.1	20.8	15.6	13.3	9.7		
	2%	DB	7.6	7.2	11.6	17.6	23.7	28.7	29.8	29.1	25.5	19.5	14.8	9.9		
		MCWB	5.7	4.7	6.6	9.7	15.6	19.9	21.7	21.5	18.1	13.8	12.2	7.7		
	5%	DB	4.8	5.4	9.0	14.9	20.9	26.4	27.9	27.5	23.2	17.5	12.8	7.6		
		MCWB	3.0	3.0	5.1	8.1	13.6	18.8	20.6	20.5	17.2	13.3	10.4	5.9		
	10%	DB	2.4	3.5	7.2	12.7	18.0	24.1	26.4	26.0	21.8	15.8	11.3	5.5		
		MCWB	0.8	1.3	4.0	6.9	11.9	17.7	19.7	19.2	16.9	12.1	9.1	3.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	7.4	10.7	14.1	18.6	23.5	24.7	24.1	21.6	17.8	14.9	11.1		
		MCDB	10.3	8.0	13.1	19.5	25.9	29.3	29.7	29.0	25.3	20.4	16.1	12.4		
	2%	WB	6.3	5.3	7.8	11.0	16.4	21.8	22.9	22.8	20.3	15.6	13.0	8.4		
		MCDB	7.3	6.9	10.0	14.8	22.4	26.5	27.7	27.1	23.4	18.4	14.4	9.1		
	5%	WB	3.2	3.2	5.8	9.3	14.7	20.3	21.8	21.7	18.9	14.3	11.3	6.3		
		MCDB	4.6	4.8	8.5	13.4	19.1	24.3	26.5	25.8	21.7	16.5	12.3	7.5		
	10%	WB	0.8	1.7	4.1	8.0	13.3	19.0	20.8	20.7	17.8	12.8	9.2	3.5		
		MCDB	2.1	3.4	6.5	11.4	17.1	22.6	24.6	24.1	20.1	14.7	10.9	5.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	9.2	10.4	10.3	10.0	10.0	10.4	10.4	10.2	8.8	8.9		
		MCDBR	10.5	10.4	12.1	14.6	14.9	13.2	12.3	12.4	12.4	12.5	10.0	11.3		
	5% WB	MCWBR	9.6	8.4	8.2	8.4	7.9	6.8	6.0	6.4	6.9	8.3	7.8	9.8		
		MCWBR	10.2	9.5	10.8	12.4	12.5	11.6	10.5	10.7	9.9	10.4	8.8	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.303	0.329	0.355	0.384	0.433	0.447	0.414	0.375	0.338	0.319	0.293		
	taud		2.435	2.403	2.431	2.413	2.376	2.253	2.230	2.311	2.403	2.504	2.522	2.507		
	Ebn at Noon		842	892	911	910	888	841	824	840	850	843	804	807		
	Edh at Noon		73	90	100	111	118	134	136	121	102	81	68	63		
(44) All-Sky Solar Radiation	RadAvg		1.63	2.52	3.69	4.60	5.17	5.55	5.82	5.28	4.25	2.72	1.74	1.29		
	RadStd		0.12	0.20	0.22	0.49	0.62	0.46	0.41	0.28	0.25	0.25	0.16	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-1.60	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	N/A

Nomenclature: See separate page